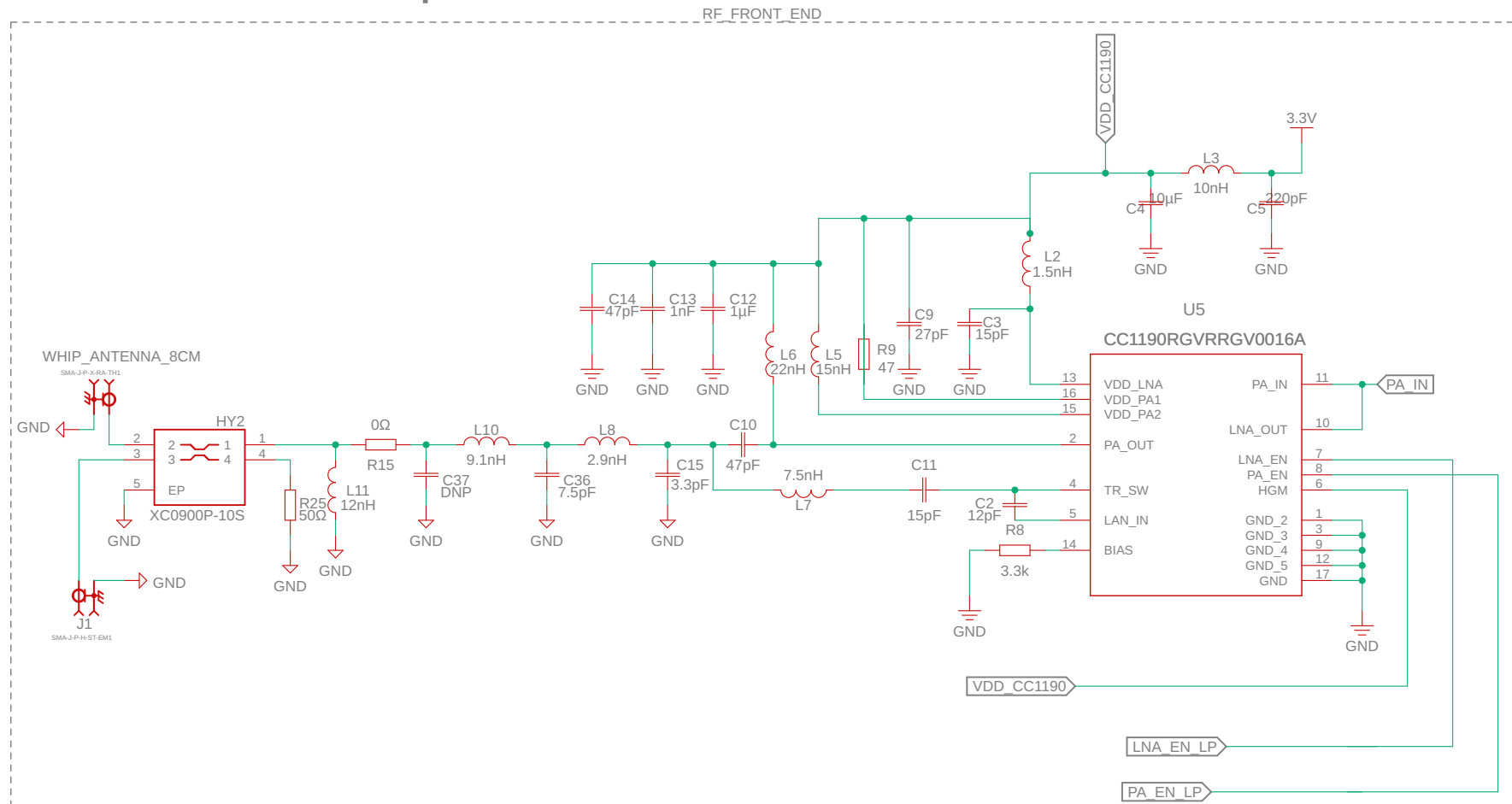


CC1190 and Whip Antenna



LNA_EN_LP and PA_EN_LP are both connections to GPIO pins on the microcontroller

VDD CC1190 is also 3.3V

WHIP_ANTENNA_8CM and J1 are SMA ports

HY2 is a 915 MHz coupler, if using different make sure to check coupling factor

C37 doesn't need to be placed

RICE ECLIPSE

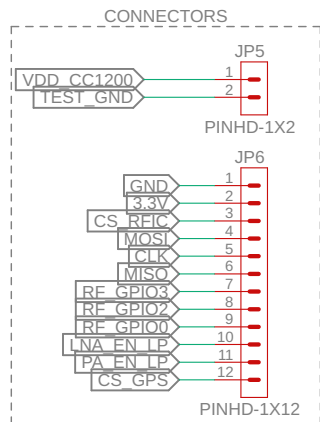
TITLE: **RAVEN V1**

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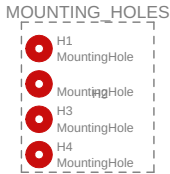
JST 12-Pin Connector and Header Pins



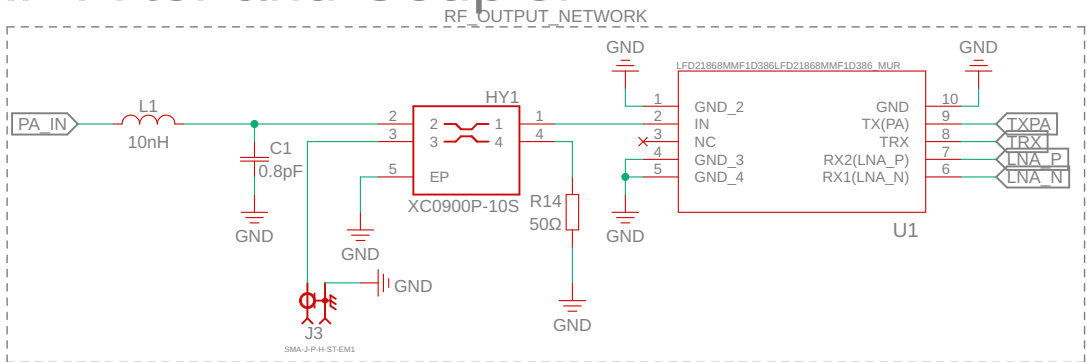
JP5 is a 2x1 header pin that serves as another test point to evaluate the power and ground, belongs near the CC1200

JP6 is the JST 12-pin Connector, it connects to a microcontroller on a separate board, a microcontroller is necessary for operation

Mounting Holes



RF Filter and Coupler

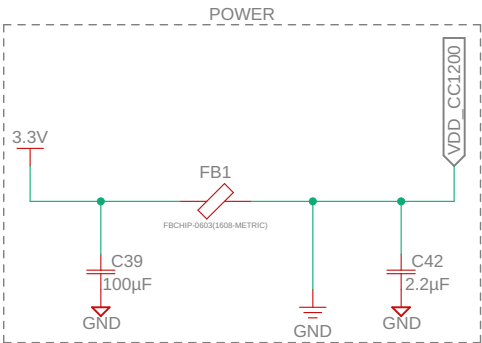


J3 is an SMA port

TXPA, TRX, LNA P, and LNA N are all RF traces

U1 is the RF Filter

CC1200 Power Filter



VDD CC1200 stays at 3.3V, but is filtered for noise

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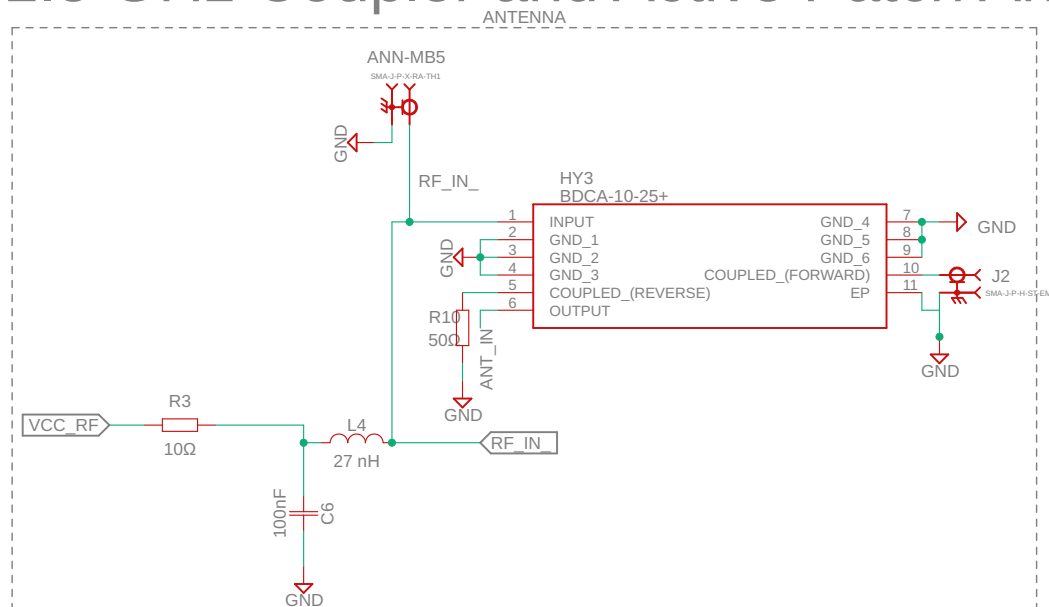
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1.5 GHz Coupler and Active Patch Antenna

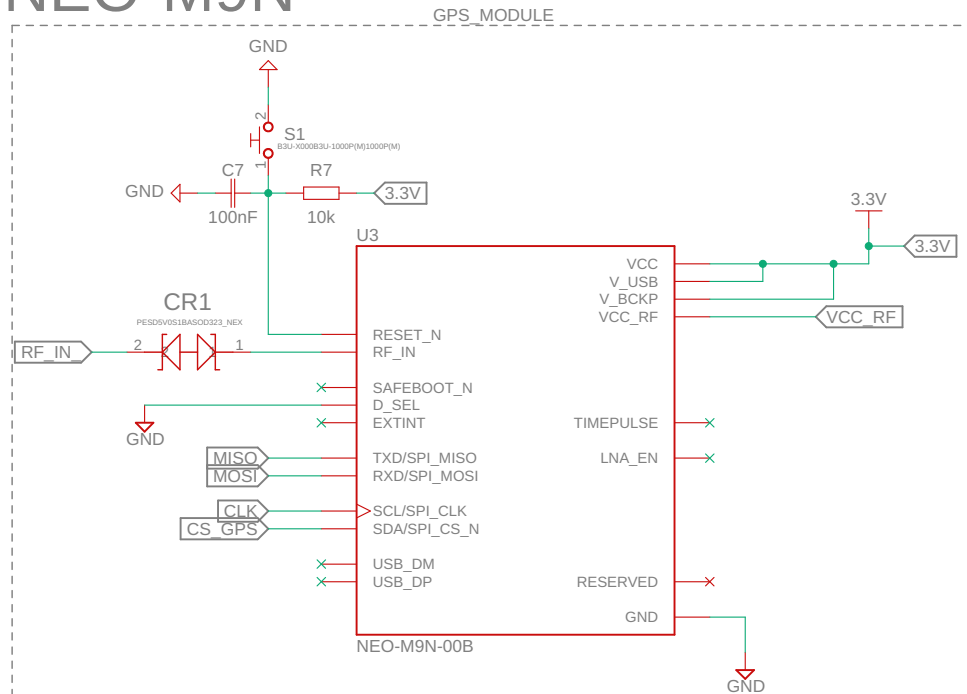


The ANNA-MB5 was chosen as the active patch antenna because it is compatible with the NEO-M9N, alternatives can be used assuming they're also compatible

ANNA-MB5 and J2 are the SMA ports

If using a different coupler remember to account for coupling factor

NEO-M9N



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